

Chapter 1

The Explanatory Vocabulary of Cognition

The truth is that science started its modern career by taking over ideas derived from the weakest side of the philosophies of Aristotle's successors. In some respects it was a happy choice. It enabled the knowledge of the seventeenth century to be formularised so far as physics and chemistry were concerned, with a completeness which has lasted to the present time. But the progress of biology and psychology has probably been checked by the uncritical assumption of half-truths. If science is not to degenerate into a medley of *ad hoc* hypothesis, it must become philosophical and must enter upon a thorough criticism of its own foundations.

A. N. Whitehead, *Science and the Modern World*, New York: The New American Library of World Literature, Inc., p. 18

Cognitive Phenomena and Folk Psychology

In science, the process of seeking to understand is called *explanation*. What distinguishes different sciences is that they seek to answer different questions and explain different phenomena. Frequently, however, different sciences address the *same* events, though in highly different ways. This would be the case, for example, if a physicist, a psychologist, and a biologist addressed the event in which a baseball pitcher throws a curve ball and attempts to say why and how the event occurred. Often, the most distinct characteristic of explanations provided by two disciplines is the vocabulary used to express both their phenomena and their explanations. This is more than a superficial question of terminology. Scientists cannot choose a vocabulary entirely for its interest and convenience, though they can (and frequently do) augment and refine the vocabulary to suit their needs. This situation exists because at least part of the vocabulary specifies the *explananda*, which carries the initial puzzle and which occurs in the substantive portion of such questions as:

Why does such and such happen?

What (hidden) things are responsible for such and such phenomenon? and

How do these things bring about that phenomenon?

Cognitive psychology is fundamentally tied to a certain class of terms which in part define the phenomena it seeks to explain (but only in part, for a consequence of explanation is that it frequently redefines its explananda) and in part dictate the sort of accounts that qualify as putative explanations. Questions of *why*, *what*, and *how* are bound so intimately to certain terms that an account based on a completely different vocabulary is simply, and correctly, dismissed as a misunderstanding of the question.

My purpose in this chapter is to introduce the reader to the idea that explanations are relative to particular vocabularies. The chapter is not intended as a contribution to the philosophy of science. I will be discussing what, at least in certain quarters, is fairly standard material but with the purpose of setting the stage for ideas to be developed in the remainder of the book. Although my goal here is not primarily to show the inadequacy or incompleteness of explanations based strictly on behaviorist ideas or neurophysiology, some discussion early in this chapter is devoted to confronting what I have discovered to be some major recurring objections to cognitive science (or “cognitivism”) from members of the psychology community. The reason I bother is that contrasting the cognitivist approach with behavioral and biological approaches is a good way to introduce cognitive science and discuss its nature. I do this chiefly by using informal, mundane examples and by playing on the reader’s intuitions. Thus, anyone familiar with the standard arguments (for example, Chomsky, 1957, Fodor, 1963b) may do well to skip to the next chapter.

The major message of this chapter is that although many of us are committed materialists, believing in physical causality and similar reasonable propositions (such as the importance of the brain in causing behavior), the problem of explanation in psychology forces us to adopt a certain kind of taxonomy of behavior. Specifically, it is because explanations attempt to capture generalizations, and different vocabularies reveal different generalizations, that we find ourselves forced to resort to what I call a *cognitive vocabulary* in revealing certain fundamental patterns of intelligent, largely rational behavior in certain creatures, and, I might add, artifacts. We will also find, perhaps not surprisingly, that the cognitive vocabulary is roughly similar to the one used by what is undoubtedly the most successful predictive scheme available for human behavior—folk psychology.

I will return to the question of choosing an appropriate descriptive

vocabulary later in this chapter; for now, an informal example will serve to illustrate this point. Suppose you are standing on a street corner and observe a sequence of events that might be described as follows. A pedestrian is walking along a sidewalk. Suddenly the pedestrian turns and starts to cross the street. At the same time, a car is traveling rapidly down the street toward the pedestrian. The driver of the car applies the brakes. The car skids and swerves over to the side of the road, hitting a pole. The pedestrian hesitates, then goes over and looks inside the car on the driver's side. He runs to a telephone booth at the corner and dials the numbers 9 and 1.

What has happened? Why did the pedestrian run to the phone booth? What will he do next, and why? If you are a determinist and a materialist, you believe that every movement that occurred in that event, no matter how small, and every change in the physical state of people and objects in that scene instantiates some natural physical law, as well as any applicable law of chemistry, biology, or neurophysiology.¹ Neglecting quantum-mechanical effects, nothing is left either to chance or to non-physical vital principles. Since, according to this view, everything that happened follows from some natural law, you might think it possible, at least in principle, to explain the entire sequence of events in this incident by recourse to the terms and laws of natural science (that is, physics, chemistry, biology, and so on).

Explaining, however, is a special activity whose success in a given case depends to great extent on how the explanation relates to the way we describe the phenomenon to be explained. A typical way to explain phenomena in natural science is to cite nomological, or universal, laws and relate them to descriptions of phenomena couched in the same vocabulary as that of the laws (or to descriptions of the phenomena defined in terms of this vocabulary). Thus physics might explain the trajectory of the car and of various objects, including people and people's limbs, in terms of their earlier states, the forces and energy transfers taking place, and the structural properties (or initial conditions) of the situation, together with the relevant physical laws.

1. Outside of quantum mechanical effects, science does seem to subscribe to the general belief in determinism, as well as to materialism, this in spite of the fact that scientists often offer opinions to the contrary. For example, in a survey of physics professors, John Macnamara (personal communication) found that 75 percent of the respondents claimed they do not believe the world is truly deterministic. Fortunately, such libertarianism appears to surface only when scientists are asked specifically to give a philosophical opinion. As far as I can determine, no one actually practices science according to that view. Invariably, theories are built to account for deterministic causes (again, outside quantum mechanics, where the situation is unclear either way). This merely demonstrates the futility of asking practicing scientists for philosophical opinions concerning the foundations of their work.

Obviously, such an explanation would provide an inadequate answer to the question, "Why did the pedestrian run to the telephone booth?" Many answers to this question are possible that would be more satisfactory:

Giving *reasons* (for example, in order to phone).

Giving *conditions* (because of the accident witnessed).

Or more systematic explanations could be given:

A rule is that, in the event of an accident, one should get help. A prerequisite for obtaining help is to communicate the request. One way to do that is by phone. A prerequisite for using a phone is to be located beside one. One way to accomplish that, if a phone is in view, is to run to it. And so on.

However, neither an answer in physical terms or in biological terms (for example, by innervating a cycle of opposing tensor muscles, which causes the position of one foot to be placed ahead of the other) would be an adequate answer to any of the *why*, *what*, or *how* questions, though they *might* form part of the overall story of how it was possible for the organism to carry out the actions mentioned in the psychological account; that is, they might form part of the explanation of the organism's *capacity* (a topic that will be treated at greater length in chapter 7).

There are general reasons why one account of a sequence of events might qualify as an explanation while another *true* account of the same sequence does not. These reasons have to do with the fact that such claims as "The occurrence of X (together with . . .) *explains* the occurrence of Y" are not, in general, equivalent (that is, they need not preserve truth values) when we replace the X or the Y by phrases that refer to exactly the same event or to the same objects (in general, "explains" provides what philosophers refer to as an "opaque context").² I will return to this point briefly, because the entire issue of the non-equivalence of neurophysiological, behavioral, cognitive, and perhaps other modes of description rests on the fact that this discussion is conducted in the context of *explanation*, where we are concerned not just with the veracity (or even the general usefulness of certain kinds of descriptions) but with the ability of different descriptions to *capture generalizations*, and hence with the adequacy of *explanations* couched in these different vocabularies.

2. I owe this point to an unpublished paper by Block and Fodor (1972). I don't know how standard the view is, but it seems correct to me. I should also point out that I am making a small point here; the more general question of *when* something counts as an explanation—particularly, to a *why* question—is enormously difficult (see, for example, Bromberger [1966]).

To return to the example of the pedestrian and the automobile, it appears—at least, intuitively—that an adequate explanation of the events in the incident must mention, among other things, that the pedestrian *perceived* the collision, *recognized* it as an event that is classified as an accident, *inferred* that there might be an injury, went over to *determine* whether anyone had in fact been injured, *deduced* from what he saw that that might be the case, *decided* to seek help based on the *knowledge* he possessed of the proper treatment of injured persons, *noticed* a telephone booth nearby, *recalled* the number for emergencies, and dialed the number with the *intention* of seeking help. While a more technical explanation might not use these particular terms, or might use more or fewer of them, it is clear that this class of vocabulary—as opposed to a physical or biological class—is precisely what appears to be needed actually to explain the *why*, *what*, and *how* of the event. These are cognitive terms, what many philosophers, following Franz Brentano's analysis of the special aspects of mental activity (see, for example, McAlister, 1976), call *intentional terms*. Although Brentano's thesis is not discussed in this book, the idea of an "intentional vocabulary" plays an important role in the discussion of psychological explanation.

It should be noted that I imply no contradiction here in suggesting that a cognitive description refers to the person following rules or doing things for a reason or reasons. It is no contradiction, as Ryle (1949) once put it, to speak of baseball players simultaneously following the laws of physics and the rules of baseball. All I am suggesting in this informal example is that some phenomena seem to demand explanations couched in a certain vocabulary.

However one views the merits of a particular explanation in the example, it at least suggests that *prima facie* constraints exist on what counts as an explanation of some phenomena. Of course we cannot let the case rest on intuitive examples alone; perhaps the intuitive criteria for the "cognitive satisfaction" of explanations are too naive; perhaps when we ask for an explanation, we do not actually want a scientific explanation, just some informal common-sense *reasons*. Perhaps all we are looking for is a story that *makes sense*. Whereas many psychologists (especially those with a biological or a behavioral orientation) seem to take this view for granted, there is reason to believe that biological or behavioral accounts are grossly inadequate for addressing certain questions because they are incapable of expressing the relevant generalizations.

What I argue in this book (primarily by example and by trying to make a plausible case, rather than by philosophical rigor, which I leave to those better equipped for the latter) is that, no matter how scientific

and technical an explanation one might wish to give, whether from a physical, a biological, or a behavioral perspective, it is still true in a most fundamental sense that if the pedestrian in our example does not *perceive* the event as an emergency, does not *know* what to do, or does not *remember* the phone number, the entire sequence could be quite different. Here, I claim that this is both true *and* relevant to our making sense of the situation. For example, it is crucial in distinguishing this sequence from one that starts off the same way in every respect (that is, with the same environmental conditions) but in which the pedestrian has, say, interpreted some feature of the situation so as to warrant the *inference* that the accident was not serious, or does not *know* what to do, or has *forgotten* the phone number. Further, I claim that any explanation which denies, say, that *being told* X (for instance, being told that the emergency phone number is 911 instead of 91) is, under specifiable conditions of motivation, a sufficient condition for *knowing* X and consequently behaving in certain predictable ways, will encounter considerable difficulty in explaining even the simplest psychological regularities—to take a random example, those of human conditioning. As Brewer (1974) shows persuasively, an account of human conditioning experiments that makes use of the notion that a subject is being *informed* of what will happen, or what he or she is expected to do, provides a better explanation of the observed phenomena than does an account based on reinforcement contingencies.

Also, I argue that a reason why this language (that is, talk of “knowing,” “perceiving as,” “being informed of,” and so on) is needed is that other sequences of behavior that are equally regular and systematic *could* be induced merely by varying the information available to the person, that is, by changing the person’s beliefs. This concern with counterfactual cases, with *what might have been*, is a hallmark of explanatory—as opposed to merely descriptive—accounts. Furthermore, it is the way the counterfactual states of affairs could be brought about that constitutes one of the features of “cognitive” systems. These points will be raised again at length in connection with the notions of cognitive penetrability and cognitive capacity that occupy much of the later chapters. All I want to do in this chapter is provide some familiar, informal examples of the phenomena by way of illustrating the point that behavior which you and I naively know to be both systematic and rational may nonetheless appear plastic in the extreme, even to the extent of seeming nondeterministic, when described using a vocabulary or taxonomy drawn from biology or behaviorism.

Capturing Generalizations

There are many reasons for maintaining that explanations of behavior

must involve cognitive terms in a way that does not serve merely as a heuristic or as something we do while waiting for the neurophysiological theories to progress. One of the principal reasons involves showing that there are regularities and generalizations which can be captured using cognitive terms that could not be captured in descriptions using behavioral or physical (neurophysiological) terms.

This sort of argument has frequently been made in the philosophical literature, and so will not be repeated here in detail. (See, however, Fodor, 1980a, 1980c, 1978a, 1968b; Putnam, 1960, 1967, 1973; Davidson, 1970; Dennett, 1978a.) I do, however, want to sketch the argument briefly and informally, because some of the principles are directly relevant to various points that arise in this book—bearing, for example, on such issues as the nature of mental processing, transduction, and mental representation. Also, I claim later on that the reasons which lead us to believe that cognitive theories must be stated in terms such as those used in the examples above—rather than in physical, neurological, or behavioral terms—are precisely those that lead us to believe that the cognitive system is correctly described as performing computations—for the reason that computation is the only detailed hypothesis available for explaining how it is possible for a physical system to exhibit regularities that must be explained as “rule following” or even as being governed by goals and beliefs.

Notice, first, that giving an account in cognitive terms is not simply to give an explanation that is more intellectually satisfying. Also, because it captures relevant generalizations, such an account is more predictive. Recall that in the example cited, we left our pedestrian in the phone booth, having dialed the digits 9 and 1. You might now ask: What will he do next? The answer is obvious when the situation is described using the particular terms I used. Because the pedestrian knows that the emergency number is 911, and because he perceives the situation to be an emergency, his next behavioral act is overwhelmingly likely to be: dial 1. The way I describe the situation, no account stated solely in behavioral terms (that is, in terms that do not incorporate the person’s knowledge or goals) can make such a prediction. The reason is, if a systematic account is to connect the prediction that the person’s next act will be to dial 1 with the “stimulus” conditions, such an account must at the very least mention: that the pedestrian *interpreted the scene as an accident*; that the pedestrian *knows or remembers* the phone number; and that the pedestrian’s behavior is an instance of the category “phone for help.” That this is true can be seen by considering the other possibilities, that is, by considering certain counterfactual possibilities. There are physically identical situations in which the prediction *does not hold* (for example, telling the person in advance that there is to be a rehearsal

of a television show on this street). And there are situations in which the prediction *does hold*—for exactly the same reason (that is, the explanation is the same) but the physical situation is vastly different (for example, instead of the person seeing the injured driver, he hears the driver's cries for help or is told by a passerby that someone has been injured). Or the physical conditions may be the same but the detailed behavior different (the person's arm is in a cast, so he asks *someone else* to dial 911, thereby realizing the prediction with highly different behavior). The point is, there are innumerable many physically distinct ways in which the same generalization can be realized; yet they remain cases of the same generalization. If that generalization were not recognized, *each instance would count as a different sequence*, and we would miss an important regularity.

It is true that such things as "beliefs" are derived from past experiences (perhaps even from experiences that were "reinforced"). Thus one might think this story could be retold using behavioral terms. The behavioral story, however, is not equivalent to the cognitive one—for an important reason. In the behavioral story past experiences must be classified in terms of a particular, *objective* taxonomy, a taxonomy that partitions classes of histories according to the physical properties of stimuli and behaviors. The way histories must be partitioned in order for them to correspond to states of knowledge, however, requires that we be capable of speaking of such things as the meaning of a sentence, or the interpretation the person placed on a certain stimulus, or the action intended by a certain behavior that formed part of the history. Unless the history leading to a particular functional state is categorized this way, we cannot use it to explain why someone does what he does.

For example, we have to classify certain events in a person's past as instances of the event type "finding out that the telephone number for emergencies is 911." If we did not classify past events in this way, we could not express a certain generalization that relates an indefinite range of antecedent histories (namely, all the different ways of finding out the telephone number for emergencies) to their common tendency to cause people to behave in certain ways, namely, to dial the digits 9-1-1. Further, without a category such as "emergency" or something similar, we could not say what was common between conditions in which people might dial 911 and those in which they are disposed to utter such exclamations as Help! or answer questions in characteristic ways.

In practice—though not in theory—the way behaviorists get around the need for such a mentalistic vocabulary is to load the description of conditions or behavior with cognitive content. Instead of a physical vocabulary, they use a "behavioral" vocabulary in which things are

described as *stimuli*, *responses*, *reinforcers*, and so on. The reason this works is that, in practice, such categories are cognitive: What serves as the functional stimulus depends on how a person interprets the situation (for example, the stimulus in the pedestrian-automobile example is *accident*; but, of course, if that person is told it is a rehearsal for a television show, the stimulus is no longer *accident* but *rehearsal* and engages the habits appropriate for that category). Similarly, what constitutes the response is also implicitly cognitive. Some particular bit of movement (accidentally bumping into a telephone while in a booth keeping out of the rain) does not count as a "response," only movements intended a certain way are counted.

The critical aspect of the connection between stimulus conditions and subsequent behavior—the aspect that must be incorporated in the theoretical account if the latter is to capture the systematicity of the person's behavior—is that it is (a) the environment or the antecedent event *as seen or interpreted by the subject*, rather than as described by physics, that is the systematic determiner of actions; and (b) actions performed with certain *intentions*, rather than behaviors as described by an objective natural science such as physics, that enter into behavioral regularities. For example, it is only under the description that the person's action is an intended telephone-dialing—rather than, say, the mere rotation of a black dial in a certain way—that the connection between the perceived stimulus situation and the action becomes systematic.

The reason for this is revealing. Even though the dial-rotating description may correspond exactly to what actually happens on a particular occasion, it does not correspond to the equivalence class of all behaviors that can be systematically related to certain previous experiences. For example, viewed thus, the description cannot be systematically related to such events as those in which the subject was told (or otherwise found out) the phone number for emergencies or those that can be systematically related to the perception of the preceding events as an accident requiring urgent assistance. The reason is, presumably, there are no valid generalizations of the form that when people know the emergency number, and when they recognize an emergency, they tend to rotate a small dial with holes in it. Such behavior as rotating a dial in a certain way is neither necessary for the generalization to hold (for example, they might use touch buttons or ask an operator), nor—even more important—sufficient. The occurrence of a certain kind of dial rotation, in which the digits 9-1-1 are rotated, is not a sufficient condition for behavior to fall under the generalization, for there are cases in which a dial might be rotated in this way without the action having anything to do with the generalization in question.

For example, a person would dial 911 as part of the general action of dialing 679-1124, or while acting in a play. That person might even dial the sequence for some thoroughly rational reason, all the while knowing the phone is out of order. These, however, are not behaviors covered by the generalization we are considering. To come under that generalization, the behavior must fall in the category of a certain intended action.

If there is any validity to the view that at least some human behavior is rational, then the systematicity of people's behavior in those cases will be stateable only when their actions are described in what I refer to as *cognitive* or *intentional* terms. On such an account the relation between conditions and actions is seen as mediated by beliefs as well as, perhaps, considerable reasoning or cognitive processing, rather than merely being linked by nomological laws. It is this remarkable degree of *stimulus-independent control* of behavior that has been the Achilles' heel of behaviorism.

The same argument can be brought to bear against attempts to predict the next behavior in the sequence from a neurophysiological or biological account. Here, I do not deny that the minute muscular movements that unfold in the instants following the point at which our story leaves off are governed precisely by physical and biological laws, and therefore are, in principle, predictable from the current physical and neurophysiological state of the person making the movement. Such a prediction, however, is both too strong and too weak a requirement to demand of our theoretical apparatus. It is too strong in the sense that it will, in principle, predict every microscopic movement that occurs, rather than predict the behavior in terms of the most systematic or regular feature that such behaviors exhibit under relevantly similar circumstances. Furthermore, to make such a strong prediction, one would need a complete description not only of the organism's current neurophysiological state but of the state of the environment as well, since that, in part, also determines the particular movements that occur.

An even more serious problem is that such a prediction is also too weak, for it does not—indeed, *cannot*—tell us that the minute sequence of movements *corresponds to the act of dialing a "one"*, which is what we really need to know in this situation. The reason this is what we need to know is not merely that it is useful or that it corresponds to our everyday way of speaking; it is that there are many situations in which we can predict correctly that a person will perform an action such as "dialing 1" although the precise way he or she does it may lie outside a uniform set of regularities covered by the theory. The sequence of movements may be *psychologically indeterminate*, inasmuch as the choice of alternative realization or alternative element of the

equivalence class is under the control of such noncognitive factors as the actual physical circumstances. In spite of this situation there may yet be an important generalization that *can* be stated in psychological terms. This is just another way of saying that, taken under a cognitive description certain kinds of behavior enter into a system of generalizations.

Another, perhaps more revealing way to cast the discussion concerning the limitations of physiological explanation is to note that the difficulty with such explanations is that they provide a distinctly different causal account linking each of a potentially unlimited number of ways of learning the emergency phone number, each of a potentially unlimited number of ways of coming to know that a state of emergency exists, to each of a potentially unlimited number of sequences of muscular movements that correspond to dialing 911. In so doing, the neurophysiological story misses the most important psychological generalization involved: regardless of how a person learns the emergency number, regardless of how he or she comes to perceive a situation as an emergency, and regardless of how the person's limbs are moved in dialing the number, a single, general principle is implicit in the entire set of these sequence. That principle can be exhibited only if we describe the events, using the kinds of terms invoked above, for example, recognizing an emergency and remembering a way to get help. Moreover, a theory's failure to exhibit such a principle is a serious shortcoming. Not only would such a theory fail to tie together various related phenomena, it would also fail to support certain relevant counterfactuals, that is, make predictions about what *would* have happened had the circumstances differed in various ways from the way they actually were. For instance, in our example such a theory would be silent on such counterfactual conditionals as what would have happened had the phone not been visible from where the pedestrian was standing; what would have happened had it been out of order; had the line been busy; or even had the phone been a Touch-Tone, as opposed to a dial phone, thus requiring different finger movements.

My point here is that phenomena can be described using a variety of terms that carve up, or taxonomize, the world in various ways. A hierarchy of type-token relationships exists among such terms. For example, there is an infinite variety of acoustical descriptions of events corresponding to the utterance of a word, an infinite variety of ways to communicate a word (spoken, typed, printed, handwritten, morse code, and so on), an infinite number of ways to tell someone something, an infinite variety of ways someone comes to know something. At each level of such a hierarchy important and interesting generalizations exist. It seems to be the case that in cognitive psychology (that is, the psy-

chology of perception, thought, problem-solving, and so forth—the psychology of intellectual processes), the interesting and important generalizations are best captured by what I call “cognitive terms”.

Cognitive science rests on the foundational assumption that there exist a natural set of generalizations that can be captured by using such cognitive terms, therefore that there is a natural scientific domain that is the subject matter of cognitive science. (Later, I will have more to say concerning the question of what characterizes this domain.) So long as such a domain exists, capturing generalizations within it is an important scientific goal. To succeed, however, we must adopt the right taxonomy or descriptive vocabulary. Another, equally important goal is to capture the generalizations in a manner that does more than merely tap our informal knowledge of the way people behave. We also want to be able to relate these generalizations to possible, physically realizable mechanisms, and that’s where computation can serve the important bridging role.

The Stimulus-Independence of Cognitive Behavior

I have claimed that an important reason for requiring a cognitive vocabulary for explaining behavior is that behavior, at least human behavior, is highly plastic in certain well-defined ways. This characteristic emerges most clearly in the degree to which behavior is stimulus-independent. Most nonbehaviorists believe that the reason behavior is stimulus-free is that what people do depends to a great extent on what they believe at the moment, how they perceive a situation at the moment, on what they think will be the consequences of various behaviors, and so on. This stimulus-independence thus is not capricious or stochastic; it is merely governed by different principles. This is an empirical, not a logical, point. As far as we know, it is not true of much of the behavior of a frog, many of whose responses appear to be the under control of the environment, as the environment is objectively described in physical terms. If we attempt to describe human behavior in terms of physical properties of the environment, we soon come to the conclusion that, except for tripping, falling, sinking, bouncing off walls, and certain autonomic reflexes, human behavior is essentially random. Yet we know that human behavior, if described in cognitive terms, is highly regular and systematic. And that, of course, is not very good news for behaviorism.

There have been attempts to capture the antecedents of cognitive regularities in physical terms, notably in the study of perception. We shall see in chapter 6 that it is important to build a bridge between

physical and cognitive terms; at that point I make some proposals about how the bridge might be built. Traditional attempts to relate perceptual regularities to physical properties, however, have encountered much the same difficulties that I alluded to. That is, the regularities that exist are to be found among perceived (cognitively described) properties or what Pike (1967) calls *emic* properties, not among objective (physically described) or *etic* properties. For example, the well-known perceptual psychologist Julian Hochberg has spent a lot of time trying to translate some of the Gestalt laws of perception into physical terms as part of what was to be a psychophysics of form perception. Despite some success in specifying "perceived complexity" in physical terms, the enterprise was abandoned. Hochberg (1968) reported that his earlier optimism had been premature; he could not specify, in physical terms, the relevant stimulus attributes that determine the way a pattern is perceived. He goes on to provide persuasive demonstrations that virtually no candidate physical properties (for example, particular physical features) are either necessary or sufficient for a person perceiving some situation in a certain way—for perceiving a stimulus *as* a something or other in the distal scene.³ Following are some examples of the sort of radical variations in the physical properties of proximal stimulation that give rise to the same perception.

A line drawing of a cube in perspective, which includes the back (potentially hidden) lines, is called a Necker cube. Except for apparent motion, which I will ignore in these illustrations, it displays the marks of a perceptual reaction referred to in note 3. Clearly the cube need not be presented as drawn lines; dashes, points, and even end points of other lines can produce the same percept. If a field is uniformly filled with random points, the same percept can be produced by arranging for the points that would lie along the lines of a superimposed Necker cube to be perceptually distinct in almost *any* way—including being a different color, size, and shape (within interesting limits—cf. Julesz, 1975) or undergoing a small synchronous movement relative

3. It should be noted that in these demonstrations Hochberg is careful to distinguish between a true perceptual product (percept) and the results of deliberate reasoning. It would be less surprising (though the same principle is involved) if it proved impossible to physically specify environmental conditions that lead eventually to the same beliefs, since clearly the same conclusions can be inferred from very different methods of presenting the information. In all cases where a claim is made that the same percept occurs, a subject not only reports the same perceived form but also reports properties that are taken as distinguishing marks of an involuntary perceptual reaction, namely, spontaneous interpretation of the two-dimensional form as a three-dimensional object, spontaneous reversals of ambiguous patterns, and spontaneous apparent motion (when a sequence is presented).

to the other points.⁴ The percept can also be produced sequentially over time by moving a screen containing a slit over any of the static forms described above (at a suitably high speed). Although one might view the latter procedure as painting the form on the retina, it turns out not to be necessary even to produce the actual spatial distribution of the pattern on the retina. The perceptual pattern can also be produced by moving the figure behind a stationary slit, thus ensuring that all stimulation is superimposed on a single line on the retina (Parks, 1965; Howard, personal communication). Various other sequential methods are possible as well. The most intriguing (and exotic) presentation, however, employs the random-dot stereogram method developed by Julesz (1971). Here, two random-dot patterns are presented, one to each eye. The patterns are identical except along the lines or regions that would be covered by a Necker-cube drawing (there, the dots of one random figure are systematically displaced relative to the other figure). Both figures are completely random patterns, but a systematic cross-correlation between the two figures carries information about the location of lines. Such a display also produces the perception of a Necker cube when—and only when—viewed stereoscopically. This exotic display can be combined with other forms of variation mentioned above.

It is not obvious what, if any, limits exist on ways to present stimulation compatible with the same perceptual interpretation. Furthermore, in each case the same regularities among perceived aspects have been found. For example, when a particular line is perceived as the top edge, the surface it bounds and the other contiguous lines are perceived as part of the top face; and when a particular face is perceived as the near face, it is perceived simultaneously as being somewhat larger than the far face. This holds true even when the figure undergoes a spontaneous reversal.

No wonder Hochberg despairs at the prospect of giving an account of perception in physical terms:

(1) because we don't have a single physical dimension of stimulation by which to define the stimulus configuration to which a subject is responding, or by which we can decide what it is that two different patterns, which both produce the same form percept, have in common . . . [and] (2) because the subject can perceive very

4. There are a few restrictions here, and a fair amount is known about the optical properties that can serve to generate the perception of lines and edges (see, for example, Julesz [1975], Marr [1982]). Nonetheless, the range of alternative physical causes of a perceived line remains unbounded.

different forms in response to the same physical stimulus, . . . which form he perceives critically determines the other answers he gives to questions about shape, motion, size, depth, and so on.

This is not to deny that *some* causal chain connects stimulation and perception. The point is simply that there exist regularities stateable over perceptual (or cognitive) categories that are not stateable over properties of the stimulation itself. This, in turn, is true because it appears that virtually no physical properties (including arbitrarily complex combinations and abstractions over such properties) are necessary and sufficient for the occurrence of certain perceptions; yet it is these perceptions that determine psychological regularities in behavior. In fact, there could not be such properties in general since, as perceptionists are well aware, the way something is perceived can vary radically with physically identical stimuli and can be the same with physically very different stimuli.

Another way to put the matter is to say that organisms can respond selectively to properties of the environment that are not specifiable physically, such properties as being beautiful, being a sentence of English, or being a chair or a shoe. These properties are not properties involved in physical laws; they are not *projectable properties*. Fodor (1984) views a system's capacity to respond selectively to nonprojectable properties as precisely what differentiates inferencing systems from systems that merely react causally to environmental stimulation. I return to this view in chapter 5 and again in chapter 7, where I take a slightly different position; for now, it suffices to remark that it is not surprising that an organism reacts to nonphysical or nonprojectable properties of the environment inasmuch as "reacting to an environment" (at least for humans, and probably for many animals) typically involves such processes as drawing inferences to the best available hypothesis about what, in fact, is out there in the distal world.

This is not to say that everything in perception involves inference (the output of transducers, discussed at length in chapter 6, does not), only that some things do. In particular, recognizing or identifying a stimulus as an instance of some category does involve inference; that's what we mean when we speak of seeing a stimulus *as* something. This recognition, in turn, is why what we see a stimulus *as* depends on what we know. And, of course, it is what we see things as that determines their effect on our behavior. When we see a certain red light *as a traffic light*, we stop—and our seeing the light as a traffic light depends on knowledge of our culture's conventions. That, in a nutshell, is why the physical properties of stimuli do not determine our behavior.

Phenomena as "Events under Descriptions"

The question, which vocabulary is the appropriate one for a particular domain, can be viewed in a slightly different way, one that helps introduce the discussion in chapter 2, in which I claim there is a need for different levels of explanation.

If an event is to be explained by some theory, it must be viewed as being nonunique. It must be seen as a sample from a large equivalence class of events among which no distinction is made by the theory. Within the domain of a theory there is an unlimited number of distinct events (in principle, distinguishable in *some* respect) that nonetheless are counted as equivalent; that is, they are, to use the linguistic term, considered to be in "free variation." Thus if what we are interested in is providing a theoretical explanation, there is no such thing as a phenomenon in some fixed or absolute sense. One cannot *point* to a phenomenon except as a rough way of distinguishing a problem domain. Any time we seek to explain a phenomenon we always do so in relation to some way of conceptualizing, or dividing, the space of possible alternatives—indeed, some way of characterizing what constitutes potential alternatives.

The particular way in which one chooses to divide a set of alternatives is revealed by the vocabulary used to describe the phenomena. Take the case of human or animal behavior. If one describes the behavior in terms of the trajectories of limbs in space and time, then two behaviors are automatically considered equivalent if they have identical trajectory descriptions, regardless of the particular muscles or neural mechanisms responsible, the goals served by the movement, the intentions realized, or its final consequences on the immediate environment. If the behavior is described physiologically, in terms of the muscles that are innervated, a wide range of trajectories and purposive actions will be treated as equivalent. On the other hand, if the behavior is described in terms of certain end results (for example, a button being depressed), as it is to some extent in operant conditioning, the description collapses over an even larger range of trajectories, muscle contractions, and goals; it does not, for instance, distinguish between the deliberate act of pressing a lever and merely bumping into it accidentally. Similarly, if behavior is described in terms of such meaningful units or actions as reaching, grasping, walking, writing, or speaking, then a far different equivalence relation is imposed, that defines what will be counted as the "same behavior."

To summarize: Different descriptive vocabularies entail different structures in the space of possible events from which some particular, observed instance is assumed to be a sample. A theory never explains

an entirely unique event, only an event viewed against a background of distinctions and equivalences defined by the vocabulary with which the events are described. That is what I mean when I say that theories address phenomena as "events under descriptions."

Two factors are relevant to the specification of this structure of phenomena, and hence to choosing a descriptive vocabulary. One is our pretheoretical notion of the domain of inquiry, or of the set of questions we take as requiring a systematic answer. The other factor is the empirical structure of the phenomena in that general domain. The informal example discussed above is intended to show the relevance of both factors. For instance, in the accident example an account can, in principle, be given solely in biological or physical terms. There are two problems with this sort of account, however. The first is that the account does not address the questions in the original domain of inquiry; it simply fails to deal with the *psychological* puzzles or with the phenomena under the intended interpretation. The second, closely related problem is that such an account does not capture the relevant regularities in the behavior.

It is an empirical fact about some behavior of humans and other animals that the regularities we are primarily interested in cannot be expressed listing certain biological and physical descriptions. This is true not only because we have various "culture-specific" notions of what regularities are "interesting" (though that in itself is enough to warrant the study of such regularities), but also because many regularities in our own behavior are clearly a result of our acting in certain ways *in order* to achieve certain results—that is, we act intentionally. Further, there must be a fairly reliable, systematic relation between the intentions we have when we act and the consequences of these acts, since such a reliable relation is precisely what is required for getting along in both social and natural environments. (For example, we need to know such things as that it is dangerous to walk off the edge of a cliff, a generalization that requires such nonphysical categories as *danger* and *walk*.) Physical and neurophysiological terms taxonomize the world in ways that do not permit us to express such generalizations. They often distinguish aspects of the world and of behavior that are equivalent with respect to their psychological import, and sometimes fail to make distinctions that are psychologically relevant. For this reason, descriptions cast in such terms typically fail to capture important psychological generalizations concerning human behavior.

Perhaps it is not surprising that strikingly different regularities emerge under different descriptions of events, and consequently that the descriptions suitable for stating the regularities of physics, chemistry, and

biology are unsuitable for stating the regularities of cognitive psychology. As Fodor (1975) puts it,

Any pair of entities, however different their physical structure, must nevertheless converge in indefinitely many of their properties. Why should there not be, among those convergent properties, some whose lawful interrelations support the generalizations of the special sciences? Why, in short, should not the kind predicates of the special sciences *cross-classify* the physical natural kinds?

Physics develops the taxonomy of its subject matter which best suits its purposes But this is not the only taxonomy which may be required if the purposes of science in general are to be served—e.g., if we are to state such true, counterfactual supporting generalizations as there are to state. So there are special sciences, with their specialized taxonomies, in the business of stating some of these generalizations. If science is to be unified, then all such taxonomies must apply *to the same things*. If physics is to be basic science, then each of these things had better be a physical thing. But it is not further required that the taxonomies which the special sciences employ must themselves reduce to the taxonomy of physics. It is not required, and it is probably not true. (p. 25)

One could, in fact, turn the entire question around and ask why the referents of such physical terms as energy are so abstract, removed from experience, and psychologically complex. The taxonomy of physics, the set of basic terms in its vocabulary, was chosen to make the statement of physical laws simple, elegant, and revealing. In a remarkable study of the relationship of geometry and sense perception, conducted in 1924, Jean Nicod (1970) makes the same point:

. . . the formation and growth of physics are dominated completely by the pursuit of simple laws or, better, of the simple expression of laws. This expression can in fact only be obtained by marking complex things by simple names. For nature is constituted in such a way that it is not the simple things that enjoy simple laws, and so, in order to simplify the laws, we must complicate the meanings of the terms. Energy, matter, object, place and time themselves, taken in the physical sense, and, more generally all the terms which are employed by physics and which do not express elementary observations, derive all their utility, their whole *raison d'être*, from this compelling need for simple and striking statements of the laws of the sensible world. (p. xviii)

The statement of cognitive regularities as symbolic processes is another

matter, however, and may require that we give "simple names" to very different "complex things."

Whatever the reason for the radical cross-classification provided by biological, physical, and cognitive taxonomies, the observation that different descriptive vocabularies are appropriate in these different domains should be clear from the example discussed above. My point here is that our pretheoretical notions of what constitutes the domain of cognitive psychology have been shaped by our experience with certain kinds of empirical regularities, so that the notions bear some *prima facie* validity, at least, as a first approximation (we see some examples in chapter 9 where these intuitions go astray as the science progresses and a more refined perspective on its boundary develops). Thus we intuitively count as psychological, regularities which, for example, relate "perceived" properties (such as the identity of objects we see in a scene) to meaningful or intentional actions (for example, "reaching for the salt"), whereas we count as at least partly non-psychological the relation between tripping over a stone and hitting one's knee. With no such pretheoretical distinction pressed upon us by various extrinsic considerations, the need for a separate science of psychology might not have arisen; the study of our behavior might simply have proceeded by treating humans as merely another physical, biological object. Among the considerations that suggest at least the first-order view of an autonomous psychological domain of inquiry, and thus argue for the importance of considering events "under a psychological description," are some of our strongest intuitions concerning the reasons for our own actions. These considerations provide plausible empirical—hence, falsifiable—hypotheses concerning the nature of regularities in human behavior. Some of them are:

(a) The intuitive presuppositions of our naive questions (for example, How do we solve problems? What happens when we read, speak, or think?), presuppositions that determine to some extent the sort of an account that will be viewed as addressing the questions on their terms. Of course, we might discover that such naive questions are poorly formulated because they have false presuppositions; but this discovery would then be a substantive discovery of some intrinsic interest. Further, it can hardly be true that all the *why* and *how* questions concerned with the aspects of behavior important to us have false presuppositions. If it were true, we would have to conclude that no science dealing with such puzzles was possible. In view of even the modest success of both artificial intelligence (which addresses some of these questions) and folk psychology, this seems unlikely.

(b) The compelling view each one of us possesses regarding why or

how we do certain things (because we are paid to, because we were taught a way to do it). While many such accounts undoubtedly turn out to be incorrect, or at least inadequate in some respect, there can be no question that *some* such accounts are correct yet they can be expressed only using certain kinds of cognitive terms to describe the environment and the behavior. In other words, there exist what appear to be clear-cut cases in which we know our behavior must be characterized over a certain nonphysical vocabulary that mentions such things as our intentions and goals. Even explaining something as mundane as why a subject in an experiment pressed the left rather than the right button must mention the subject's intent in pressing the button (for example, to indicate which of two stimuli were perceived, or to indicate agreement versus disagreement), which, in turn, depends on the subject's interpretation of the task, and so on. Thus we have to view the behavior in terms of how the subject intended it or in terms of an intentional description of the behavior.

(c) The fact that a certain rough taxonomy of the world and of other people's actions results in a reasonably successful and predictive folk psychology. While it is likely that many, perhaps most, hypothetical mechanisms that folk psychology appeals to are impotent to provide adequate explanations of behavior (since, for example, homuncular explanations abound in folk psychology), the fact that the predictions made concerning the everyday social behavior of people are so often correct strongly suggests that we have made some of the correct cuts through phenomena (that is, that the taxonomy or descriptive vocabulary over which our beliefs are expressed is more or less the right one).

(d) The rationality of much (though not all) of our behavior, which can be exhibited only if we have a taxonomy that allows us to relate beliefs and goals to intended actions. Rationality itself cannot be applied to descriptions cast in terms of biology or physical trajectories, yet it is one of the properties of at least some of our activity that must somehow enter into an explanation of our behavior; at least some explanations of our behavior take the form: Person *S* had goal *G* and beliefs *B*₁, *B*₂, *B*₃, . . . , *B*_{*n*}, one or more of which had the content that action *A* could help achieve goal *G*, and hence, on the assumption that *S* acts rationally, *that is why S* carried out a certain behavior as an instance of *A*. Obviously, much remains to be said about rationality, especially since there are large areas in which people do not appear to behave in accordance with normative principles of rationality (cf. Nisbett and Ross, 1980; Tversky and Kahneman, 1974; but see Kyburg, 1983). Nonetheless, the principle of rationality—suitably embedded within a system of

limited, imperfect mechanisms—is indispensable giving an account of human behavior, for reasons explored in chapter 2.

These considerations provide us with a first-draft specification of a domain of inquiry, or to put it as I did above, they specify what it is to view phenomena *under a cognitive description*. I have suggested that lawlike generalizations or regularities in behavior of special concern to cognitive psychology are likely to be captured only if the behavior and its antecedent conditions are described in terms of such a taxonomy. I have also claimed that such a taxonomy is distinct from that provided by other disciplines, especially physics and biology, and that the taxonomy typically is not only more abstract than that of these sciences but that it classifies events in equivalence classes whose boundaries typically do not coincide with the boundaries of classifications based on the other sciences. So far, though, I have said nothing that is not also true of all the “special sciences,” that is, of every science other than basic physics. In each case, generalizations from each science are stateable only over their own special vocabulary; consequently, the lawlike generalizations of these sciences are not reduceable to some finite combination of physical laws. Each category over which regularities are stated in a special science stands in a type-token relation to categories of physics, which means a category in, say, economics cannot be reduced to a finite disjunction of categories of physics. I will have more to say about type-token hierarchies in chapter 2, inasmuch as I argue that there is more than one principled level of this hierarchy within cognitive science itself.

What I suggest is that there is an important difference between the generalizations that belong to the “cognitive sciences” and those that occur in physical or other “natural” sciences, because, in cognition, there is another type-token level that does not apply in natural-science accounts that is relevant to the notion of rationality, as well as to what Brentano first described using the term *intentionality*, a property he took to be the mark of the mental. I do not discuss the issue of intentionality in its full philosophical complexity (there is an enormous technical, philosophical literature on the subject); I do examine one aspect of intentionality because it is closely related to the notion of *representation*, a notion which plays a fundamental role in cognitive explanation. Chapter 2 is devoted to a discussion of this topic. I might just note in anticipation of later discussion, however, that the need to appeal to representations in providing explanations of cognitive phenomena constitutes one of the main reasons why cognition is fundamentally a *computational* process.